

| Week    |                              | Content                                                                                                         | Reading                                                                                                      | HW Assigned                        | PA Assigned                        |
|---------|------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------|------------------------------------|
| week 1  | Tuesday, September 1, 2026   | Course policies, introduction to Th. Of Comp.<br><i>First look at Finite State Automata (FSA)</i>               | Chapter 0                                                                                                    |                                    |                                    |
|         | Thursday, September 3, 2026  | Terminology and formal definition of FSAs<br>state diagrams, regular languages (RL)<br><i>operations on RLs</i> | Chapter 1.1                                                                                                  | PS1 Out<br>Due Thu, 09/17, 11:59pm |                                    |
| week 2  | Tuesday, September 8, 2026   | Closure of RLs under operations of union<br>and intersection,<br>the product machine                            | Chapter 1.1                                                                                                  |                                    | PA1 Out<br>Wed, 10/07, 11:59pm     |
|         | Thursday, September 10, 2026 | Non-deterministic Finite Automata,<br>closure under concatenation & Kleene star                                 | Chapter 1.2                                                                                                  |                                    |                                    |
| week 3  | Tuesday, September 15, 2026  | Other regular operations<br>Equivalence of NFAs and DFAs,<br>Subset Construction                                | Chapter 1.2                                                                                                  |                                    |                                    |
|         | Thursday, September 17, 2026 | NFAs whose DFAs have exponential size,<br>Regular Expressions                                                   | Hopcroft, et al. 2.3.6<br>Chapter 1.3                                                                        | PS2 Out<br>Due Thu, 10/01, 11:59pm |                                    |
| week 4  | Tuesday, September 22, 2026  | Conversion from REs to NFAs/DFAs<br>GNFAs & conversion from DFAs to Res                                         | Chapter 1.3                                                                                                  |                                    |                                    |
|         | Thursday, September 24, 2026 | Distinguishable & equivalent states in a DFA,<br>Distinguishability Lemma, the Quotient<br>automaton            | Hopcroft, et al. 4.4                                                                                         |                                    |                                    |
| week 5  | Tuesday, September 29, 2026  | State minimization algorithm for DFAs,<br>the Pumping Lemma for regular languages                               | Hopcroft, et al. 4.4.3<br>Sipser 1.4                                                                         |                                    |                                    |
|         | Thursday, October 1, 2026    | Examples of non-regular languages, using the<br>the Pumping Lemma to prove non-regularity                       | Chapter 1.4                                                                                                  | PS3 Out<br>Due Thu, 10/15, 11:59pm |                                    |
| week 6  | Tuesday, October 6, 2026     | Context Free Grammars, Leftmost Derivations<br>Parse Trees                                                      | Chapter 2.1                                                                                                  |                                    | PA2 Out<br>Due Thu, 11/05, 11:59pm |
|         | Thursday, October 8, 2026    | CFG Ambiguity, CFL Closure Properties,<br>Chomsky's normal form,                                                | Chapter 2.1                                                                                                  |                                    |                                    |
| week 7  | Tuesday, October 13, 2026    | Monday Schedule -- no class                                                                                     |                                                                                                              |                                    |                                    |
|         | Thursday, October 15, 2026   | Nondeterministic Pushdown Automata & CFLs                                                                       | Chapter 2.2                                                                                                  | PS4 Out<br>Due Thu, 10/29, 11:59pm |                                    |
| week 8  | Tuesday, October 20, 2026    | Review                                                                                                          |                                                                                                              |                                    |                                    |
|         | Thursday, October 22, 2026   | MIDTERM EXAM                                                                                                    |                                                                                                              |                                    |                                    |
| week 9  | Tuesday, October 27, 2026    | Equivalence of NPDA's and CFGs                                                                                  | Chapter 2.2                                                                                                  |                                    |                                    |
|         | Thursday, October 29, 2026   | The Pumping Lemma for CFLs<br>Examples of non-CFL languages                                                     | Chapter 2.3                                                                                                  | PS5 Out<br>Due Thu, 11/12, 11:59pm |                                    |
| week 10 | Tuesday, November 3, 2026    | Turing Machines                                                                                                 | Chapter 3.1                                                                                                  |                                    | PA3 Out<br>Due Fri, 12/04, 11:59pm |
|         | Thursday, November 5, 2026   | Multi-tape Machines & Their 1-Tape Simulators                                                                   | Chapter 3.2                                                                                                  |                                    |                                    |
| week 11 | Tuesday, November 10, 2026   | Nondeterministic Turing Machines and Their<br>Deterministic Simulators                                          | Chapter 3.2                                                                                                  |                                    |                                    |
|         | Thursday, November 12, 2026  | The Universal Machine, Decidable Languages                                                                      | Chapter 4.1                                                                                                  | PS6 Out<br>Due Tue, 11/24, 11:59pm |                                    |
| week 12 | Tuesday, November 17, 2026   | Undecidable & Unrecognizable Languages                                                                          | Chapter 4.2                                                                                                  |                                    |                                    |
|         | Thursday, November 19, 2026  | Additional Undecidable Problems<br>Complexity of Turing Machines                                                | Chapter 4.2, 7.1                                                                                             |                                    |                                    |
| week 13 | Tuesday, November 24, 2026   | Classes P and NP                                                                                                | Chapter 7.2, 7.3<br>(definition of class NP<br>is from recommended<br>textbook, page 431,<br>Section 10.1.3) | PS7 Out<br>Due Mon, 12/07, 11:59pm |                                    |
|         | Thursday, November 26, 2026  | Thanksgiving!                                                                                                   |                                                                                                              |                                    |                                    |
| week 14 | Tuesday, December 1, 2026    | Polynomial Reductions                                                                                           | Chapter 7.3, 7.4                                                                                             |                                    | PA3 Due<br>Due Fri, 12/04, 11:59pm |
|         | Thursday, December 3, 2026   | Cook's Theorem, NP-Complete Problems                                                                            | Chapter 7.4                                                                                                  |                                    |                                    |
| week 15 | Tuesday, December 8, 2026    | Review                                                                                                          |                                                                                                              |                                    |                                    |
|         | Thursday, December 10, 2026  | FINAL EXAM                                                                                                      |                                                                                                              | PS7 Due<br>Due Mon, 12/07, 11:59pm |                                    |